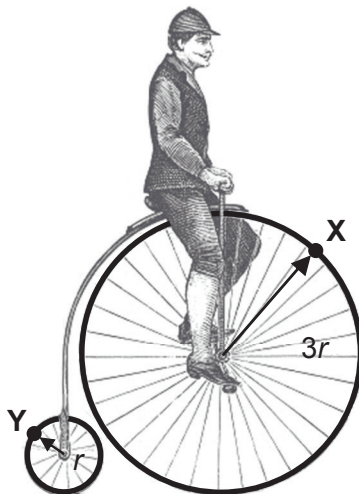


PAPER 1 / QUESTION 86

25 April 2024 / TZ2 (file label) / HL Paper 1 / original question 19 / 2224-6707

86. An old-fashioned bicycle is moving with a constant speed v . The front wheel has a radius 3 times that of the rear wheel. Points X and Y are positioned on the front and rear wheel as shown.



What is $\frac{\text{acceleration of X}}{\text{acceleration of Y}}$?

- A. $\frac{1}{9}$
- B. $\frac{1}{3}$
- C. 3
- D. 9